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Automated Detection and Severity Assessment of Dysarthria using Raw Speech

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***Abstract*—Dysarthria is a medical condition that impairs an individual's ability to speak clearly due to muscle weakness or paralysis. To diagnose and monitor dysarthria severity, this article proposes the use of a deep learning model that utilizes raw speech waveforms. This approach eliminates the need for feature engineering and enhances the model's ability to handle noise and speech variability. The proposed system was compared to a standard convolutional neural network (CNN) model and was found to perform better in both dysarthria severity classification and dysarthria/healthy control classification tasks. The results indicate that the SincNet model achieved an accuracy of 95.7% and 99.6% in these tasks, respectively. The proposed system can aid clinicians in diagnosing and monitoring dysarthria severity and could have broader applications in speech-related disorders. The study emphasizes the importance of using raw waveform-based models for speech analysis and demonstrates the effectiveness of SincNet in automatic dysarthria/healthy control classification and dysarthria severity assessment.**

***Index Terms*—Dysarthria severity level assessment, Speech-related disorders, Deep learning, SincNet, Raw waveforms.**

I. Introduction

Effective communication through speech is a fundamental aspect of human interaction that can significantly impact one's quality of life. However, some individuals may experience difficulty with speech production due to neurological causes, resulting in motor speech disorders such as dysarthria [1]. Dysarthria is characterized by weakness, paralysis, or lack of coordination of the motor-speech system, leading to reduced intelligibility, audibility, naturalness, and efficiency of vocal communication [2], [3]. In recent years, there has been an increased interest in developing automated methods for the detection and severity assessment of dysarthria using raw speech, which could potentially improve the diagnosis and treatment of this disorder [4]. Raw waveform models in the classification of dysarthric speech offer several advantages. Raw waveforms capture a more complete representation of the speech signal, including subtle variations in time and frequency, which can help distinguish between healthy and dysarthric speech [5]. By eliminating the need for manual feature extraction, raw waveform models save time and reduce the risk of biased feature selection. Additionally, these models allow for the use of end-to-end approaches, such as convolutional neural networks (CNNs), which can learn complex representations directly from the raw waveform, leading to improved classification accuracy [6]. Lastly, raw waveform models offer a more straightforward and practical approach for dysarthria assessment as they require minimal preprocessing of the speech signal, making them easier to use in clinical settings.
The main contributions of the article are fourfold:

- Developed an automated system that can detect dysarthria by analyzing raw speech waveforms.
- Proposed a method for assessing the severity of dysarthria based on raw speech waveforms and achieved high accuracy in predicting the severity level.
- Employed a specific type of neural network architecture called SincNet for dysarthria detection, which has not been used in this context before as par of our knowledge.
- Demonstrated the effectiveness of using SincNet for dysarthria detection and assessment.

The subsequent sections of the article are arranged as: Section II presents the motivation and the existing literature on dysarthria detection. Section III describes the dataset and proposed methodology used in the study. Section IV reports the experimental results of dysarthria detection and severity assessment using the proposed methodology, showing high accuracy in both tasks. Finally, Section V concludes the key findings of

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the suggested work and discusses their implications for future research in the field.

II. Motivation & Related Work

Automated detection and severity assessment of dysarthria using raw speech is a challenging task in the field of speech processing and healthcare [7]. Dysarthria is a speech disorder that affects a person's ability to articulate words clearly due to weakness or paralysis of the muscles involved in speech [8]. Accurate detection and assessment of dysarthria can aid in the diagnosis and treatment of various neurological disorders that can cause dysarthria, such as Parkinson's disease, stroke, or cerebral palsy [9]–[11]. Raw waveform modeling is a promising approach for automated detection and severity assessment of dysarthria [6] because it allows for a more comprehensive representation of the speech signal compared to traditional feature-based approaches. By using raw waveforms, the model can capture more detailed information about the temporal and spectral characteristics of the speech signal, which are essential for accurately detecting and assessing dysarthria. However, recent advancements in technology have enabled the development of automatic assessment methods using deep learning frameworks. A fresh way of measuring dysarthria severity using sound descriptors has been implemented [12]. They classify speech into different severity levels within the UA-Speech corpus and the TORGO database. Further, the same author proposes a new approach based on machine learning to classify dysarthric speech as either intelligible or non-intelligible using bidirectional long-short term memory networks [13]. Additionally, they proposed a transfer learning technique that utilizes a pre-trained acoustic models results in better handling of channel noise. The transfer learning approach demonstrated a significant improvement of roughly 6%, in contrast to traditional machine learning methods.

A recent study has introduced a new way to measure the severity of dysarthria by analyzing the way people speak. The researchers collected data from people with and without dysarthria who spoke English and Korean. They looked at aspects of speech like pitch, speaking speed, and rhythm and used them to train different computer algorithms to categorize the speech as having different levels of dysarthria severity [14]. Moreover, another study assesses various architectural choices, with two types of speech features: Mel-frequency cepstral coefficients (MFCCs) and constant-Q cepstral coefficients (CQCC). The UA-Speech and TORGO databases were used for evaluation, with the CNN model achieving the highest accuracy of 93.2% and 96.18%, respectively. The investigation compared various neural network models for automatic dysarthria severity level classification [15]. Additionally, another study was conducted by the author, who had previously utilized different DNNs. The evaluated models consisted of deep squeeze and excitation (SE) CNN models with residual blocks, a shallow CNN, and a convolutional recurrent neural network (CRNN). SE-CNN model showed significant improvement, achieving a 10% margin over the baselines in the speaker-independent scenario [16].

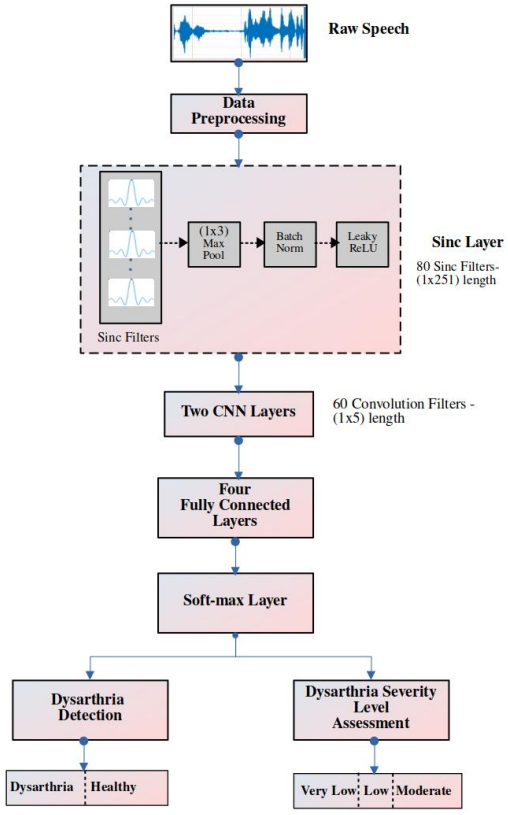


Fig. 1: Flow chart of automated dysarthria detection and dysarthria severity level assessment using SincNet

III. Proposed Methodology

A. Raw Waveform Modeling

One of the key advantages of SincNet is its ability to model raw waveforms directly. Traditionally, speech signals have been processed using hand-crafted features such as MFCCs [17]–[20] or perceptual linear predictive (PLP) features or wavelet coefficients [21]. While these

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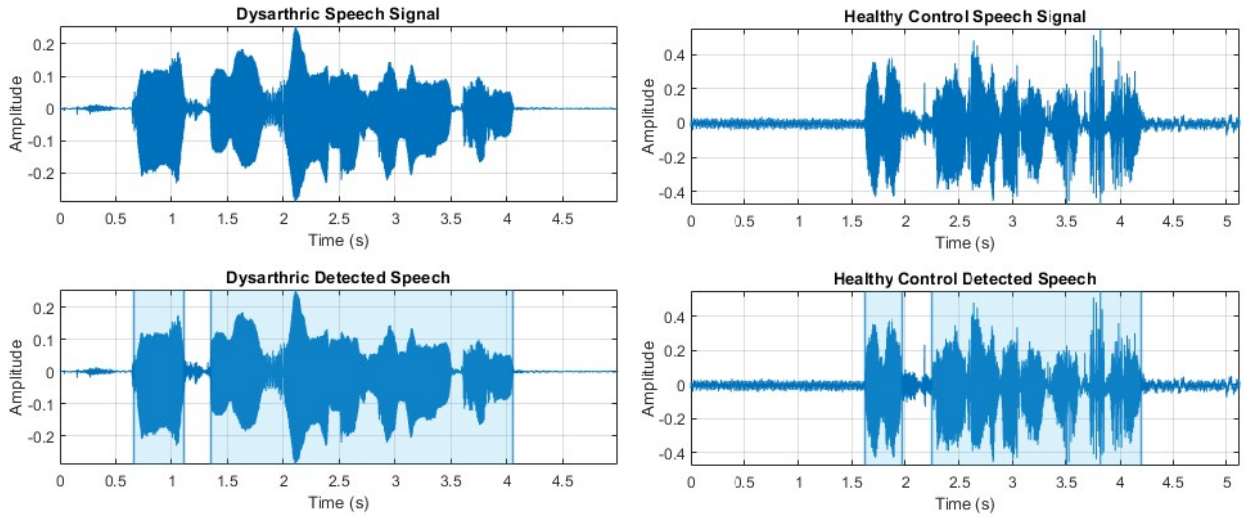


Fig. 2: Comparison of dysarthric and healthy control speech of utterance "*You wish to know all about my grandfather*" and corresponding detected speech from raw audio

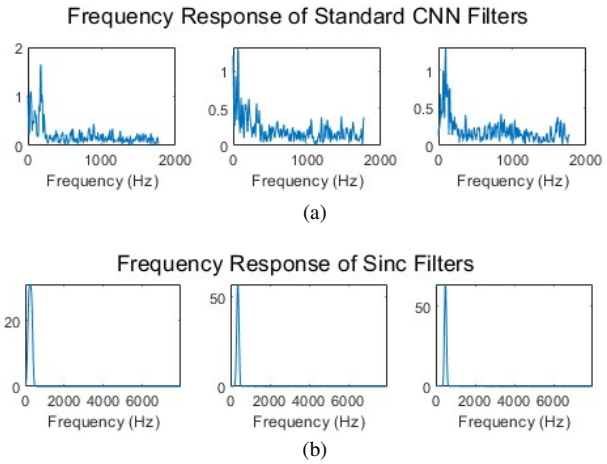


Fig. 3: Frequency responses of filters learned from (a) Standard CNN & (b) SincNet at different audio frames

features have proven effective in many applications, they are often limited in their ability to capture the full complexity of the raw waveform. In contrast, SincNet operates directly on the raw waveform, allowing it to capture both the fine-grained and global structure of the signal. This is accomplished through the use of the sinc-layer, which applies a set of adjustable band-pass filters to the input signal. These filters are designed to capture different frequency components of the signal, allowing SincNet to analyze the signal at multiple scales. The use of raw waveform modeling has several advantages. First, it eliminates the need for hand-crafted features, which can be time-consuming and difficult to optimize. Second, it allows for greater flexibility in modeling the input signal, since the network can learn to extract the most relevant features directly from the raw waveform. However, it allows for more accurate modeling of speech disorders, since these disorders often manifest in subtle variations in the raw waveform that may be difficult to capture with hand-crafted features.

B. SincNet

The proposed paper makes use of the SincNet neural network architecture as depicted in Fig. 1, which was designed mainly to evaluate raw speech signals [22]. For diagnosing speech abnormalities like dysarthria, this makes it an optimal preference. SincNet is designed to evaluate raw speech signals and has multiple layers, including a special sinc-layer that uses band-pass filters with configurable center frequencies and bandwidths to analyze the input signal in detail. This is in contrast to standard CNN filters, which are not consistent with perceptual information as shown in Fig. 3(a). The Sinc-Net layer attempts to learn from only few paramters i.e., upper and lower cut-off frequencies of the band-pass filter, as shown in Fig. 3(b). Then the input signal is processed through two 2D convolutional layers with 60 filters each to extract local features. Fully connected layers with batch normalization and leaky ReLU activation are then used to extract high-level features from these local patterns. Four fully connected layers are employed, each with 256 hidden units. The output is then fed into a softmax layer to make predictions on

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TABLE I: Details of TORGO dataset in the detection of dysarthria and severity level assessment

Details of Dataset	Detection of Dysarthria (Binary-Class)		Dysarthria Severity Level Assessment (Multi-Class)		
	Dysarthria	Healthy Control	Very Low	Low	Moderate
No. of Speakers	8	7	3	2	3
Gender Composition	3F/5M	3F/4M	2F/1M	1F/1M	3M
Speaker IDs	F01, F03, F04, M01, M02, M03, M04, M05	FC01, FC03, FC04, MC01, MC02, MC03, MC04, MC05	F03, F04, M03	F01, M05	M01, M02, M04
No. of Raw Audio Samples	1000	1000	525	202	273

different labels, including whether or not the speaker has dysarthria and the severity level of the disorder. Finally, the softmax layer outputs posterior probabilities over the different labels, which are aggregated to make an overall prediction on the target label.

IV. Experimental Results

This section aimed to evaluate the effectiveness of an automated detection and severity assessment system for dysarthria using raw speech data through two experiments. The first experiment aimed to classify dysarthric speech from healthy control speech (binary class), while the second experiment aimed to classify the severity level of dysarthria into three classes: very low, low, and moderate (multi-class). Two different network types, including a standard convolutional neural network (CNN) and a SincNet, were trained and evaluated for both experiments of dysarthria detection. The TORGO dataset was split into a training set (80%) and a test set (20%). After being trained, both the standard CNN and SincNet models were evaluated to classify new or unseen speech signals as either dysarthric or healthy control. The performance of the classifiers was evaluated using overall accuracy and relative improvement (RI) metrics.

TABLE II: Automated dysarthria and severity level detection in terms of overall accuracy (%) and relative improvement, RI (%) of two different network types

Classification	Standard CNN	SincNet	RI (%)
Dysarthria Detection (Binary Class)	89.8	99.69	11.69
Severity Level Assessment (Multi Class)	82.01	95.76	16.76

A. Dataset

The TORGO database [23] is a popular database for investigating dysarthria, a speech disorder that affects articulation. It contains audio recordings and detailed measurements of speech movements from individuals with dysarthria and a control group of healthy individuals. The database includes data from 8 dysarthric speakers and 7 healthy control speakers, all aged between 16 to 50 years as shown in Table I. The recordings in the database provide rich information on speech production, including both sound data and detailed measurements of speech movements. Fig. 2 shows that individuals with dysarthria struggle to produce speech due to a diminished ability to control their articulatory movements. This is due to existing issues like prosodic abnormalities and resonance changes, which can cause slurred or mumbled speech. On the other hand, healthy individuals do not experience these difficulties and are able to produce clear speech.

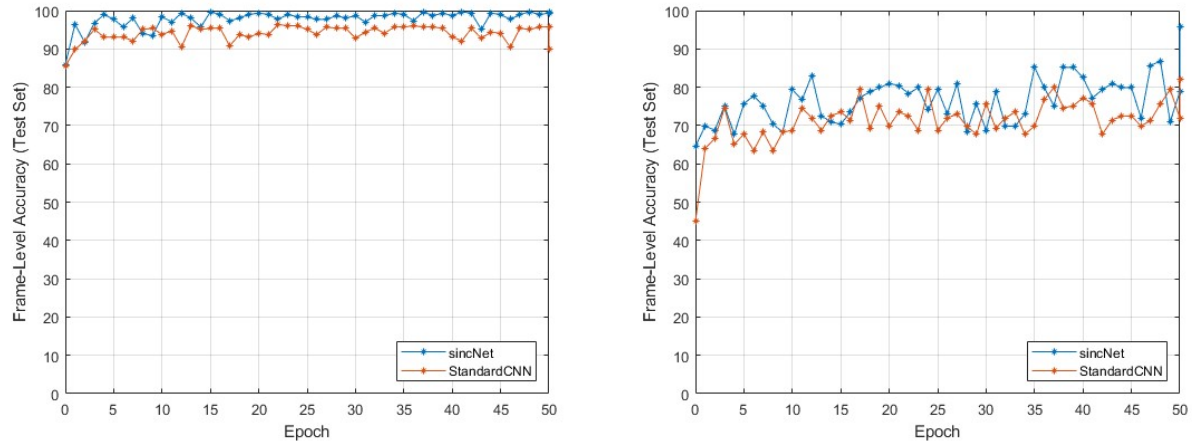
B. Data Preprocessing

The raw speech waveforms underwent data preprocessing to improve the quality of the audio signals and extract useful features for further analysis. A threshold-based approach was employed to remove silence regions and detect speech regions as shown in Fig. 2. Silence regions were identified by setting a threshold level below which any amplitude values were considered as silence, and any regions with amplitude values below this threshold were removed. The detected speech signals were segmented into frames of 200ms duration with an overlap duration of 40ms. This allowed for the analysis of the speech signal in smaller time intervals, aiding the identification of specific speech features and patterns within each frame. To avoid loss of important information at the boundaries between frames, overlapping frames of 40ms were used. This ensured that all relevant information was captured.

C. Automatic Detection of Dysarthria

The aim of this section was to evaluate the performance of the automated dysarthria detection system. To achieve this goal, the dataset, which consisted of speech signals from individuals with dysarthria and healthy controls, underwent preprocessing to remove silence regions. Subsequently, the remaining speech signals were segmented into frames of 200ms with an overlap duration of 40ms. Subsequently, the segmented raw features were utilized to train the standard CNN and SincNet models, both of which were used for binary classification, with

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(a) (b)

Fig. 4: (a) Automated dysarthria detection and (b) Dysarthria severity level detection in terms of frame level accuracy (%) for two different network types

dysarthric speech signals classified as positive subjects and healthy control speech signals classified as negative subjects.

The standard CNN model demonstrated an accuracy of 89.8% in dysarthria detection. However, by using the SincNet model, there was a significant improvement in classification accuracy, with an increase to 99.69%, is as shown in Fig. 4(a). This considerable improvement could be attributed to the SincNet model's fewer parameters and more meaningful filters, resulting in a relative improvement of 11.69%, as shown in Table II. Additionally, the findings presented in Table III indicate that dysarthria subjects experienced only 0.6% misclassification, demonstrating the efficacy of the proposed automated dysarthria detection system.

TABLE III: Confusion matrix of binary classification of dysarthria and healthy control subjects in terms of frame level accuracy (%) using SincNet

		Predicted Class	
		Dysarthria	Healthy Control
Actual Class	Dysarthria	99.4	0.6
	Healthy Control	0	100

TABLE IV: Confusion matrix of multi-class classification of the severity level of dysarthria in terms of frame level accuracy (%) using SincNet

		Predicted Class		
		Very Low	Low	Moderate
Actual Class	Very Low	99.2	0.8	0
	Low	6.8	93.2	0
	Moderate	21.1	0	78.9

D. Automatic Severity Level Assessment

Automated severity level assessment has the potential to provide a more objective and efficient method for evaluating dysarthria severity. The aim of this section was to evaluate the performance of the automated severity level assessment, which consisted of very low, low, and moderate subjects. To achieve this aim, the dataset was split into train and test sets for each class and trained using the standard CNN and SincNet models. A frame level accuracy of the test dataset is shown in Fig. 4(b) for dysarthria severity level assessment. The standard CNN model achieved an accuracy of 82.01% in classifying the severity level, which is lower than the SincNet model's accuracy of 95.76% and results in a relative improvement of 16.76%. The low accuracy of the standard CNN model can be attributed to its limited ability to learn complex features from raw waveforms and its tendency to overfit the training data. On the other hand, the SincNet model's accuracy improvement can be attributed to its fewer parameters and more meaningful filters, enabling it to better capture relevant features from the raw waveforms. This remarkable improvement demonstrates the potential of using the SincNet model for automated severity level assessment in dysarthria. Moreover, the confusion matrix presented in Table IV indicates that the automated severity level assessment system can easily identify subjects with very low severity level who exhibit long pauses and speech difficulties with only a 0.8% misclassification rate. However, in the case of subjects with low and moderate severity levels, the majority of their speech frames are close to the very

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TABLE V: Comparison of proposed model performance with previous approaches

Year/ Author	Dataset	Front-End Approach	Model	Performance
2020 Millet et al. [6]	TORGO	Raw waveforms	Per Channel Energy Normalization, LSTM and Attention model	Highest accuracy of 82.4% using time-domain filterbanks
2020 Hernandez et al. [14]	TORGO	MFCCs with voice quality or prosody	RF, SVM, NN	Highest accuracy of 75.63% using MFCC and prosodic features trained on NN
2022 Schu et al. [24]	TORGO	Handcrafted features like MFCCs, Mel spectrogram, Sparsity-based features.	Speech representation learning (SRL) + Mel spectrogram	High classification accuracy ranging from 71.1% to 100.0%
2022 Joshy et al. [15]	TORGO	MFCC, CQCC	DNN models like CNN, GRU, LSTM	Highest accuracy of 96.18% was achieved by CNN
2023 Joshy et al. [16]	UA-Speech	Mel-spectrograms	Squeeze & Excitation (SE) CNN	97.58% for speaker-dependent 57.23% for speaker independent
Proposed Method	TORGO	Raw waveforms	CNN, SincNet	99.4% for dysarthria detection and 95.76% for severity level assessment using SincNet

low severity level. This leads to misclassification rates of 6.8% and 21.1%, respectively. Moderate and low level severity are close to the very low level because they share some common speech characteristics with very low severity, such as long pauses, slow speech rate, reduced articulatory precision, and distorted prosody. These similarities can make it difficult for the automated severity level assessment system to distinguish between them, leading to a higher misclassification rate for subjects with moderate and low severity levels.

E. Comparative Analysis of Proposed System with Earlier Work

The section presents a comparative analysis of the proposed system with earlier work on dysarthria detection and severity level assessment. Different methods have been used, including time-domain filterbanks and LSTM with attention models, MFCCs with prosodic features, speech representation learning with handcrafted features, and DNN models like CNN, GRU, and LSTM with MFCC and CQCC features. The proposed method uses raw waveforms and employs two deep learning models, namely standard CNN and SincNet, and achieved an accuracy of 99.4% for dysarthria detection and 95.76% for severity level assessment on the TORGO dataset, outperforming the existing methods. This demonstrates the efficacy of the approach for dysarthria detection and severity level assessment as shown in Table V.

V. Conclusion

This article proposed an innovative and promising method for automated detection and assessment of dysarthria using raw speech waveforms. The proposed system was able to accurately distinguish between healthy individuals and those with dysarthria, and the method for severity assessment showed impressive accuracy in predicting the severity level. Moreover, the study highlights the effectiveness of SincNet, a unique type of neural network architecture, for dysarthria detection and assessment. The results of this study could significantly improve the diagnosis and treatment of dysarthria, particularly in early detection and severity assessment. To ensure the efficacy of this approach, further research could be conducted on larger and more diverse datasets. This method could be integrated into clinical practice and evaluated in a real-world setting to assess its clinical feasibility and impact.

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